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Developmental Changes in Picture Recognition

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Recognition of complex pictures by children in the first, third, and fifth grades (ages 7, 9, and 11 years, respectively) was tested and then compared with adult performance. Pictures were either of organized scenes or of collections of objects. By systematically varying the distractors in the recognition tests, recognition of four types of information was measured: inventory, descriptive, spatial relation and spatial composition information. Patterns of performance on the various types of distractors as well as overall accuracy were measured. For organized scenes, there was a gradual increase in accuracy with age but patterns of performance were similar at all grade levels. For unorganized pictures, children were much less accurate than adults and showed different patterns of responses. Children were particularly poor at processing spatial relation information from unorganized pictures. It was suggested that children as young as the first grade have developed scene schemata and use these cognitive structures to encode and retrieve organized scenes in the same fashion as do adults. However, children at least up to the fifth grade have difficulty in organizing materials structured in unfamiliar ways. The data were found to be congruent with other studies of both visual recognition and verbal recall, implicating unfamiliarity of stimulus materials as a major contributor to developmental trends in memory.

The goal of this research is to discover the kinds of information people encode and remember from complex pictures and to determine whether there are developmental changes in the kinds of information processed. Even relatively simple line drawings of scenes contain a vast amount of information, only some of which is encoded. Picture recognition, like recognition of linguistic material, is not unlimited, as can easily be demonstrated by controlling the similarity of the distractors in a recognition test. By increasing similarity, it is possible to make recognition performance decrease to chance levels, even when the differences between target and distractor are perceptually discriminable.

We can take advantage of this fact and systematically vary the dimensions on which distractors differ from targets. Using an analogy from

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signal detection theory, this technique divides the "noise" distribution into clearly defined categories, so one can determine which kinds of "noise" can be distinguished from the signal and which cannot. We have found that this technique enables us to determine exactly which aspects of pictures are encoded and retained (Mandler & Johnson, 1976; Mandler & Ritchey, 1977). Failure to recognize a particular change indicates either that that piece of information in the picture was not encoded in the first place or that the person cannot find an effective enough retrieval cue to recover it.

The technique also allows a sensitive assessment of developmental changes in recognition. Several studies of picture recognition have shown only small differences in accuracy between children and adults (Brown & Campione, 1972; Corsini, Jacobus, & Leonard, 1969). These findings have led a number of investigators to suggest that picture recognition may be relatively insensitive to developmental trends, at least from preschool years onwards (Brown, 1975; Hagen, Jongeward, & Kail, 1975). The rationale for the hypothesis is the following. Developmental changes in memory are most pronounced when mnemonic strategies are required (Flavell, 1970). Recognition, as opposed to recall, does not require such strategies and therefore smaller differences in performance between children and adults should be expected (Brown, 1975; Hagen et al., 1975).

As Brown pointed out, the data on which this hypothesis rests were contaminated by ceiling effects. Recognition was tested by randomly chosen distractors and was excellent for everyone, thus masking possible developmental differences. When distractors more similar to the targets are used, developmental differences are uncovered (Dirks & Neisser, 1977; Mandler & Stein, 1974; Tversky & Teiffer, 1976). As for the role of mnemonic strategies, it is important to keep in mind the distinction between strategies of the deliberate planful variety and the more general notion of encoding and retrieval operations, also often called "strategies." Brown (1975) divides memory tasks into "strategic" and "nonstrategic" ones, and suggests that recognition of unrelated pictures is a nonstrategic task. However, recognition of even simple materials has been shown to involve retrieval operations (Mandler, 1972). When recognition of complex stimuli (such as pictures of scenes) is required, both encoding and retrieval operations must play an important role. To the extent that these operations change with age, we can expect development differences in picture recognition (Nelson & Kosslyn, 1976).

The use of a strategy–no strategy distinction implies the difference between intentional and incidental memory, but this distinction seems less important than that between processing familiar and unfamiliar materials. A set of materials is familiar (not only individual items but relationships among them) if it matches an already acquired schema. We use the term "schemata" to refer to internal structures, built up through experience, that organize and give meaning to incoming information. Thus, in predicting and assessing developmental trends in memory, it is useful to consider the nature of the schemata that are used for encoding and storage by subjects of different ages.

Experiments on complex picture recognition require a great deal of information to be encoded in a limited period of time; since the pictures typically represent meaningful objects and complex relationships among them, the way the material is stored and retrieved can be expected to depend on the kinds of schemata that have been developed. To the extent that children have already acquired scene schemata containing the same kinds of information as those of adults, we would expect their *patterns* of responding to be similar on recognition tests that vary the types of information changed in the distractors. For example, if a schema includes information about expected spatial relations among objects in a scene, but not about the general appearance of the objects, then children, like adults, should recognize changes in the former kind of information better than changes in the latter. Only if different kinds of information are encoded by children and adults would pattern differences in recognition be expected. However, overall *levels* of accuracy should improve with age even on recognition tasks, since there are major developmental trends in the speed, exhaustiveness, and efficiency of extraction of information from visual stimuli (Day, 1975).

To the extent that pictures are *not* meaningful (i.e., do not fit familiar object or scene schemata) we would expect developmental trends to be more pronounced. Imposing meaning on nonmeaningful material seems to be a peculiarly adult characteristic. This tendency is undoubtedly more pronounced in intentional learning situations; although adults may routinely organize unfamiliar input in terms of more familiar schemata even in situations in which they are not trying to remember. If children do not or cannot organize unusual stimuli, developmental differences in levels of accuracy should increase when unfamiliar materials are processed, regardless of intention to learn. It is not obvious that even in this case different *patterns* of recognition of various types of information would be found. Different patterns might occur if an inability to relate unfamiliar material to previously acquired schemata results in distortion of normal encoding processes, e.g., lessened attention to some kinds of information.

The present experiment investigated the kinds of information encoded and retained from complex pictures by children in the first, third, and fifth grades. The data were compared with adult performance on the same pictures, reported in Mandler and Ritchey (1977). The method is based on previous studies from our laboratory. Briefly, it consists of presenting subjects with a series of complex line drawings. Half of the subjects see organized pictures (i.e., pictures of objects forming a real-world scene). The other half see unorganized versions of the same pictures (i.e., versions in which the objects are arranged in haphazard fashion). The objects themselves are equally meaningful in the two versions, but in the organized pictures the relationships among the objects are meaningful as well.

Following presentation of the target pictures, each subject is given a recognition test consisting of a series of old and new pictures. The new pictures consist of a set of eight types of transformations on the target pictures. Each transformation tests recognition of a particular type of information (Mandler & Johnson, 1976; Mandler & Ritchey, 1977). Four types of information were investigated: (a) *Inventory information,* specifying what objects a picture contains; (b) *Descriptive information,* specifying the figurative detail of the objects in the inventory (i.e., what the objects look like); (c) *Spatial relation information,* specifying where the objects are located, including their relations to other objects, such as "left of", "facing"; and (d) *Spatial composition information,* specifying areas of filled and empty space in the overall composition of the picture, without regard to the nature of the items filling spaces. The taxonomy does not include actions or reference to what is inferred to be happening in a picture, but emphasizes the more static, figurative characteristics of scenes. By using both organized and unorganized pictures and by classifying children's recognition errors according to this taxonomy, we were able to investigate pattern differences as well as differences in accuracy in recognition of familiar and unfamiliar materials.

METHOD

Subjects

A total of 144 children participated in the experiment: 48 first graders ranging in age from 6.6 to 7.6 years, mean = 7.0; 48 third graders ranging in age from 8.5 to 9.7 years, mean = 8.11; and 48 fifth graders ranging in age from 10.6 to 11.11 years, mean = 11.0. The children were randomly selected from a large public school in San Diego. Each group consisted of approximately equal numbers of boys and girls.

Stimuli

The stimuli were the same as those used by Mandler and Ritchey (1977). Two sets of 20 × 25 cm photographs of black and white drawings were used as target pictures. Each set consisted of eight pictures. The pictures in the organized set represented naturalistic, real-world scenes. Each scene contained six non-overlapping objects plus a single perspective line.[1] Four of the pictures were outdoor scenes and four were indoor scenes; within this division two of each type included people and the other two did not.

The set of unorganized pictures was produced by removing the

[1] This set of pictures was noticeably less dense in appearance than the eight object pictures used by Mandler and Johnson (1976) and Mandler and Parker (1976).

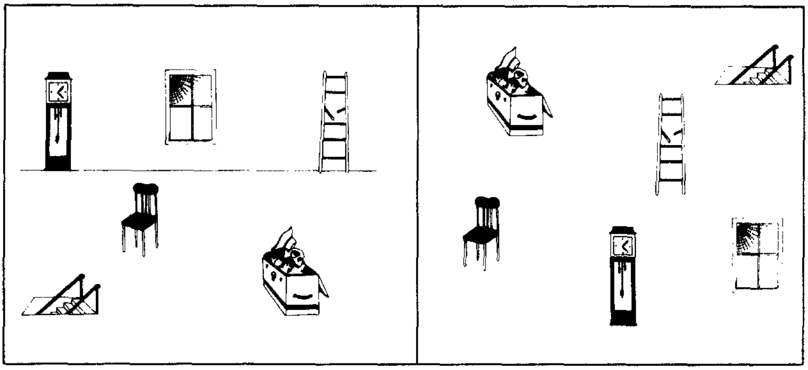

FIG. 1. An example of an organized and unorganized version of a picture.

perspective line from each organized picture, inverting the picture, and then rotating each object back to its normal upright position. Then four of the six objects were interchanged to create essentially a zero correlation between horizontal and vertical coordinates of objects in the two versions. This method of disorganizing pictures disrupted meaningful relations among objects while maintaining approximately the same density of objects in both versions of a given picture. An example of an organized and unorganized version of a picture is shown in Fig. 1.

Eight transformations were generated for each picture to create distractors in the recognition tests. One transformation tested recognition of descriptive information. This was a *Token change,* in which one object was replaced by another object of the same size, shape, and conceptual class, but which differed in details of appearance. The Token change for the attic scene in Fig. 1 was substitution of another chair for the one shown. Three transformations tested recognition of inventory information. One was a *Type change,* in which an object was replaced by a conceptually different object of the same size and shape. In the attic scene, a picture was substituted for the window. An *Addition* and a *Deletion* also tested inventory information. In the Addition to the attic scene, a sewing machine was added to the picture; in the Deletion, the chair was removed.

Additions and Deletions also test spatial composition information, since they change the areas of filled and blank spaces. Another transformation testing spatial composition was a *Move change,* in which an object was moved in the horizontal plane by a distance equal to its width. In the attic scene, the trunk was moved. A *Size change* was also used, in which an object was increased by 67% in area or decreased by 40% (so that a 67% increase would be required to match its original size). In the attic scene, the window was made smaller. We originally assumed (Mandler & Ritchey, 1977) that size changes affected both spatial composition and spatial relation information. However, unless the changes are large enough to distort the scale of the pictures, spatial relation information is not affected. The size changes of the magnitude used, although discriminable, do not noticeably change appropriate relationships among objects and thus are classed as a spatial composition change. Spatial relation information was tested by two transformations: a *Rearrangement* and an *Orientation change*. In the Rearrangement transformation two objects of approximately the same size and shape were interchanged. All Rearrangements were made along the horizontal rather than the vertical dimension so that apparent size would not be altered in the organized scenes. In the attic picture, the clock and the ladder were interchanged. In an Orientation change, an object was reversed so that it faced in the opposite direction. In the attic picture the stairs were reversed.

Transformations were chosen so that they did not violate real-world relationships or change the meaning of any scene. No object in a picture was used in more than two transformations. Transformations were assigned approximately equally often to large and small objects, to each location, and to animate or inanimate objects. The same objects were transformed for the organized and unorganized versions of a given picture.

For a practice session, an organized and unorganized version of two additional target pictures were generated. Four of the eight transformations were used as distractors for one of the pictures and the remaining four transformations were used as distractors for the other. Pictures were presented at a slight angle adjusted to each child's line of vision and placed approximately 25 cm from the edge of the table at which the child was seated.

Design

The between-subject factors were organization of the pictures (two levels), grade (three levels), and exposure time (10 or 20 sec).[2] Each cell of the design contained 12 children. Each child saw eight targets and was tested on recognition of all eight transformations. A complete recognition test of eight pictures with eight transformations on each consists of 64 targets and 64 distractors if probability of old and new stimuli is to be kept at .50. To reduce learning effects during the course of the recognition test and to avoid tiring the children, each child saw only half of the items (each target four times and each type of transformation four times). Thus, it took two children to generate a complete set of "within-subject" data points. Children were randomly assigned to pairs and their data combined for purposes of statistical analysis, in effect reducing the number in each cell of the design from 12 to 6 subjects.

[2] Exposure time was varied because at the outset it was not certain that the 10 sec exposure time that Mandler and Ritchey (1977) used for their adult subjects would be adequate for the first graders. However, 10 sec exposure produced sufficiently good recognition scores so that comparisons between the two experiments could be made.

Six orders of presentation of the eight targets were used, one for each pair of subjects in a cell. For any given order of presentation, the recognition test was arranged so that seven different pictures always occurred between successive versions of a given picture. The various types of transformations were scattered throughout the recognition test so that equal numbers of each type appeared in each quarter of testing. No more than three transformations or three targets appeared in sequence.

Procedure

Children were tested individually in sessions that lasted about 20 min. In order not to bias their processing of the target pictures toward any particular type of information, they were told only that they would see eight pictures for a short amount of time and that they should study them very carefully so they would remember everything. The eight target pictures were then presented. Next, children were informed of the nature of the changes on the pictures that would occur during the recognition test. This was accomplished by means of a brief training session. Such instruction is necessary because of the similarity of distractors to targets; even adults, unless informed, will exhibit a strong bias to say "same" to most stimuli in the early portions of the recognition test. The children were shown two new pictures, followed by four targets and four transformations for each. They were told that if a picture was exactly the same as one of the two target pictures, they should respond "same," but if any picture was a little bit different they were to respond "different." Each "different" response was either confirmed or corrected by pointing out the transformation and comparing it with the target picture. Incorrect "same" responses were corrected in the same manner.

At the end of the training session, children were reminded about the first eight pictures they had seen. They were told that they would see a group of pictures, and that half would be exactly the same as the eight pictures previously seen and half would be a little different. Children were then asked to tell how they might be different; the experimenter reminded them if they omitted any transformation. Children were then given the recognition test. They were shown the test pictures for 10 sec each. If after 10 sec no response had been given, a plain piece of paper was placed over the picture until the child responded. With rare exceptions, children responded within 10 sec. After every eight pictures for the first 24 pictures in the recognition test, children were reminded that they had to remember the first eight target pictures they had seen.

RESULTS

To facilitate comparison with other studies, the mean proportion correct scores for target pictures (i.e., hit rate, or saying "same" to targets) and for

TABLE 1

PROPORTION CORRECT ON TARGETS AND TRANSFORMATIONS FOR ORGANIZED (O) AND UNORGANIZED (U) PICTURES FOR THREE GRADE LEVELS

	Grade level					
	First grade		Third grade		Fifth grade	
	O	U	O	U	O	U
Targets	.79	.77	.80	.74	.79	.70
Type change	.71	.77	.82	.71	.90	.85
Addition	.69	.73	.87	.86	.84	.92
Deletion	.59	.60	.63	.75	.70	.67
Rearrangement	.74	.59	.93	.61	.94	.75
Orientation change	.48	.41	.63	.40	.64	.38
Token change	.44	.47	.51	.54	.61	.64
Move	.29	.29	.41	.45	.37	.47
Size change	.22	.29	.20	.31	.19	.33

the eight types of transformation (i.e., correct rejections or saying "different" to the distractors) are provided in Table 1. However, all statistical analyses used d′ scores to control for variation in the tendency to say "same" or "different" among the various conditions.[3] An analysis of variance was carried out with exposure time, grade level, and organization of the pictures as between-subject variables, and transformations as a within-subject variable. Not surprisingly, 20-sec exposure resulted in more accurate recognition (mean d′ = 1.23) than 10 sec (mean d′ = 0.90) [$F(1,60) = 12.75, p < .01$]. However, the difference in accuracy was not large and this factor did not interact with any other variable.

The main findings are illustrated in Fig. 2, which shows mean d′ scores for each transformation on organized and unorganized pictures in the three grades. There were large differences in accuracy on the various transformations [$F(7,420) = 114.37$, $p < .001$]. There was also overall improvement across grade levels [$F(2,60) = 4.66, p < .02$], but grade level interacted with type of transformation [$F(14,420) = 2.05$, $p < .02$]. A breakdown of this interaction by Newman–Keul tests showed that there was major improvement across the three grades on three transformations: Addition, Type, and Rearrangement ($p < .01$ in all three cases). Slight

[3] Each d′ score entered into the analyses was based on the hit rate for each pair of subjects, as described in the Method section, and the false alarm rate for a given transformation for that pair of subjects. In the several experiments in this series we have consistently found d′ to be more sensitive than the nonparametric A′ measure. Confidence rating data from adults indicate normal variance, with slightly greater variance in the distractor distribution than in the target distribution (cf. Mandler & Ritchey, 1977, for details).

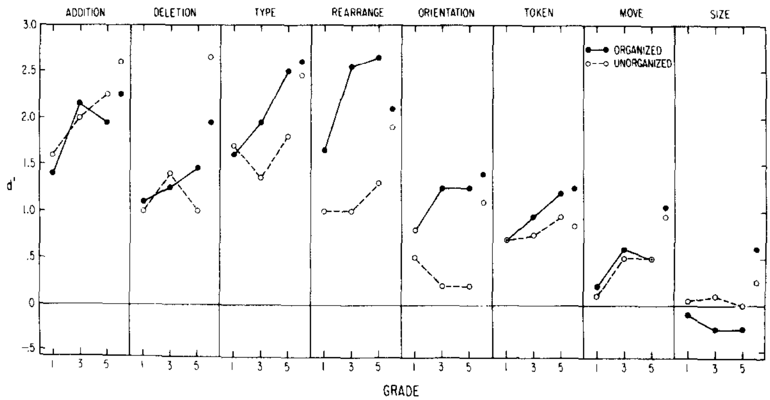

FIG. 2. The d′ scores for recognition of eight types of transformations on organized (filled circles) and unorganized pictures (open circles) in grades one, three, and five. Data from the 10 and 20 sec exposure conditions have been combined. Adult data for 10 sec exposure, taken from Mandler and Ritchey (1977) are shown to the right of the children's data for each transformation.

improvement occurred for the Move and Token changes ($p < .05$). There was no age trend for Deletions or Size changes.

Accuracy of performance on the eight transformations was ordered quite consistently across the three grades. Furthermore, with the exception of recognition of deleted objects (discussed below), the ordering of the children's performance in all three grades was similar to adult performance (cf. Mandler & Ritchey, 1977). Adult d′ scores for immediate recognition, taken from the Mandler and Ritchey study, have been presented in Fig. 2 to allow a comparison of patterning of responses, but it should be noted that the adult scores were for 10 sec exposure only and therefore somewhat underestimate the size of the child–adult differences in accuracy.

The conclusion that children and adults show the same patterning of responses is qualified, however, when the organization of the pictures is considered. Overall, children were much more accurate on organized than on unorganized pictures [$F(1,60) = 10.16$, $p < .01$]. This finding is consistent with our prior study of children's picture recognition (Mandler & Stein, 1974), and contrasts with our studies of adults. In none of the adult immediate recognition tests (Mandler & Johnson, 1976; Mandler & Parker, 1976; Mandler & Ritchey, 1977) have adults shown overall superior recognition of organized pictures.

There was an interaction between organization and transformations [$F(7,420) = 12.89$, $p < .001$]. Simple main effects analyses showed that recognition was significantly superior for organized pictures only for the Type, Rearrangement, and Orientation transformations, ($p < .01$ in all three cases). Differences in detection of Rearrangement and Orientation changes in the two types of picture were particularly large (see Fig. 2).

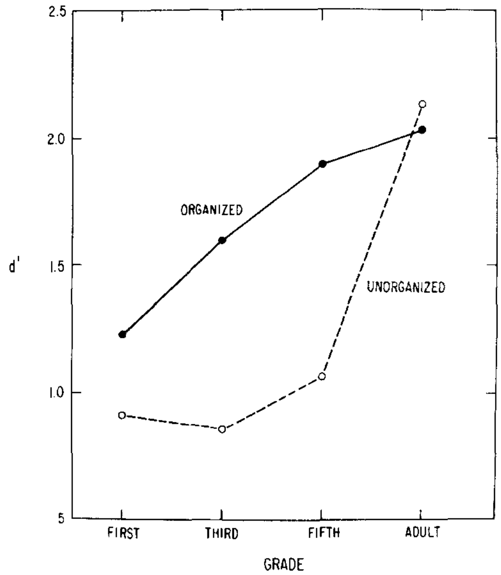

FIG. 3. Recognition of organized and unorganized pictures by children and adults in the 10 sec exposure condition.

These transformations assess spatial relation information, and the results indicate that when pictures conform to a scene schema, encoding of spatial relations is enhanced; in contrast, spatial relations tend to be ignored in pictures consisting of collections of objects. Although this trend has also been found for adults, it was much more pronounced in the children's data. The interaction of organization and grade level was not significant, indicating that the relationships just described were similar for the three grades, although the differences between organized and unorganized pictures were smaller for the first grade than for the third and fifth grades.

An analysis of variance was also carried out to compare performance on the first half vs the second half of the recognition test. Performance improved from the first to second half of testing, from a mean d′ score of 1.01 to 1.21 [$F(1,60) = 5.12, p < .05$], but this factor did not interact with any other variable. Subjects were picking up more information about the target pictures during the course of testing but did not do so differentially as a function of age or condition.

Statistical comparison of children's and adult performance could be made in the case of the 10 sec exposure condition since the adults in the immediate recognition condition in the Mandler and Ritchey study were given identical treatment. Because we were particularly interested in comparing the developmental effects of stimulus organization on performance, we excluded three transformations from this comparison, the Size and Move changes, which were too close to chance responding to show effects of organization, and the Token change, which has been unrelated to organizational factors in all our previous studies.An analysis of variance of the remaining five transformations was carried out, varying grade level (first, third, fifth and adult) and organization.

There were large main effects of both organization [$F(1,40) = 11.93$, $p < .001$], and grade level [$F(3,40) = 11.98$, $p < .001$], as well as an interaction between the two factors [$F(3,40) = 2.90$, $p < .05$]. The interaction is shown in Fig. 3. It can be seen that only the adults handled the two kinds of pictures equally well. Children had greater difficulty with unorganized pictures; although the difference between the two kinds of pictures was significant only for the third and fifth grades ($p < .01$ in both cases). There was no significant change in scores for unorganized pictures from the first to the fifth grade, and a large improvement from fifth grade to adult ($p < .001$). In contrast, there was an increase in performance on organized pictures from the first to the fifth grade ($p < .05$) and little change thereafter.

The organization of the pictures also interacted with transformations [$F(4,160) = 7.55, p < .001$]. As found in the main analysis of the children's data reported earlier, the largest differences due to organization in the 10 sec exposure condition occurred on Rearrangement and Orientation transformations. For organized pictures, children performed unusually well on these two transformations compared to adults (i.e., no significant age trends were found). For unorganized pictures, however, there was little change in accuracy of recognition of Rearrangement and Orientation changes from the first to the fifth grade, and a significant increase between fifth grade and adult ($p < .05$).

A breakdown of the interaction between grade level and transformations [$F(12,160) = 1.86, p < .05$], indicated that the most difficult transformation for children in comparison to adults was Deletion. There was no significant improvement in recognition of Deletions for *either* type of picture from the first to the fifth grade, and a large increase in accuracy from fifth grade to adult ($p < .01$). The poor recognition of Deletions is of particular interest because it contrasts with the high level of performance on the other two changes which assess inventory information, Type changes and Additions (see Fig. 2). The differences among the three transformations suggest that inventory information was indeed encoded but that recognition of Deletions requires more retrieval skills than the other two. To recognize Type changes and Additions, subjects can use the presented inventory to check for matches or mismatches; but to recognize that an object had been deleted, subjects must rely on an internally generated inventory.

The other basis on which a Deletion might be recognized is to use spatial composition information. However, Mandler and Ritchey (1977) suggested that the present set of pictures, containing fewer objects and more blank spaces than previously used pictures, tended to reduce the role of spatial composition in mediating recognition. The data from the present study are consistent with that assumption. In particular, it has been found that adults recognize changes in spatial composition better in unorganized pictures; yet for children in the present study, there was no effect of organization on any of the transformations which affect this type of information (Move, Size, Addition, and Deletion). Further, recognition of Size changes was at chance and Move changes scarcely above chance, suggesting that the children were relatively insensitive to the spatial composition of the pictures.

A final analysis was carried out to compare the four grade levels in the 10 sec exposure condition on the three remaining transformations: Token, Move, and Size changes. As expected there was no effect of organization. Grade level was a significant source of variance [$F(3,40) = 7.16, p < .001$]. Analysis of this main effect indicated that the three groups of children did not differ from each other but all were significantly less accurate than the adults ($p < .01$). Although the grade by transformation interaction did not reach significance, it was broken down by the Newman–Keuls procedure so that age trends for the three transformations could be compared. Children at all grades were significantly poorer on the Move and Size changes than the adults ($p < .05$); but there was no significant age trend in recognition of the Token transformation.

In summary, in comparison to adults, children encoded a good deal of inventory information from both organized and unorganized pictures. They encoded spatial relation information well from organized pictures but relatively poorly from unorganized pictures. Their encoding of spatial composition information from both types of picture was poor in comparison to adults. They also encoded relatively little descriptive information from both types of picture, but their performance was not notably worse in this regard than that of adults.[4]

DISCUSSION

From first grade to adulthood patterns of responding to various kinds of transformations on organized scenes were found to be highly similar. There was a gradual increase in accuracy on most transformations but, with the

[4] It is difficult to compare the present findings with the recent study of Dirks and Neisser (1977), who studied recognition of Additions and Move changes and recall of Deletions, but used a very different procedure. Comparisons are hampered by the fact that they had no single measure such as d′ to take account of false alarms as well as correct responses. Since their procedure generated some considerable developmental differences in response biases, it is especially difficult to compare the developmental trends in the two studies. However, inspection of their two separate measures suggests good recognition of Additions and poor recognition of Moves, with recall of Deletions falling in between, a pattern similar to the present findings.

exception of spatial composition information, little indication that the children processed different kinds of information than adults. Our previous studies of adults have shown that scene schemata influence encoding of spatial relation information more strongly than any other. The children's excellent performance in this regard suggests that picture recognition is controlled by such schemata at least from the first grade.

Much greater developmental differences were found for unorganized pictures. Not only was the children's performance markedly poorer but there were differences in the kinds of information encoded. In particular, spatial relation information was poorly processed in comparison to adults. The much greater effect of organization found in this study in comparison to our previous studies of adults suggests that children's memory is highly dependent upon the activation of familiar schemata; and that their recognition will be most like adults' when they are asked to remember meaningful materials structured in familiar ways.

This conclusion is not restricted to pictures of scenes. Similar findings were reported by Mandler and Day (1975) for recognition of drawings of single objects. They found no differences from second grade to adulthood in accuracy of recognizing left–right orientation of single objects, as long as the objects were familiar.[5] For unfamiliar forms, there was a much greater difference, with improvement in recognition, occurring even after the sixth grade. A plot of recognition scores on familiar and unfamiliar shapes from kindergarten to adulthood is similar to the curves shown in Fig. 3 for organized and unorganized pictures. These differences were the same in both intentional and incidental learning conditions.

There is, of course, evidence from other areas that children's memory is more dependent on familiar schemata than is that of adults. Children's serial recall of well-structured stories is similar to that of adults (Mandler & Johnson, 1977). However, children are less able to maintain accurate serial recall of stories presented in unorganized form. Mandler (in press) found that children's recall was distorted toward an ideal story structure, whereas adults could better maintain the actual input order. Similar findings have been reported by Brown and Murphy (1975). In terms of quantity of recall, Mandler's story data also look very much like Fig. 3 in the present report. By the sixth grade, children recalled as much as adults from well structured stories, but for unorganized stories, there was little improvement from the second to the sixth grade and a marked increase in recall between sixth grade and adult.

A similar interpretation can be made of studies of recall of lists of categorized words. Analyses of clustering during recall show that adults reorganize randomly presented lists into familiar categories. Children do

[5] Vogel (1977), however, found improvement from the second to fourth grades, using somewhat different materials.

not accomplish such reorganization when the structure of the materials is not obvious to them at the time of presentation. When helped to uncover the structure of the materials through techniques such as blocking (Cole, Frankel, & Sharp, 1971) or sorting (Worden, 1974), both recall and clustering become more similar to that of adults, although quantitative differences remain.

These findings suggest a somewhat different view than Brown's (1975) of developmental changes in memory. She proposes a two-way classification of tasks into "strategic–nonstrategic" and "semantic–episodic", and suggests that the smallest developmental trends will occur on "nonstrategic episodic" tasks such as picture recognition. We have avoided the distinction between episodic and semantic memory because it does not appear that the two terms can be adequately separated, perhaps especially in the kinds of recognition tasks discussed here. Both qualitative and quantitative differences in recognition were found between organized and unorganized pictures; but one would not term the former a semantic task and the latter an episodic one. Either a "semantic" or an "episodic" task can involve familiar or unfamiliar material and the latter distinction seems to be the more important one.

Further, both familiar and unfamiliar materials require encoding and retrieval strategies. However, we should distinguish between strategies requiring planful organizing activity and those in which cognitive structures are automatically accessed. When materials are presented in formats which conform to already acquired and well organized schemata, such as those for real-world scenes or stories, children as young as first graders encode the same kinds of information as adults and their memory performance is similar in kind, although not usually in quantity. A different picture emerges for recognition of unfamiliar material. The generally poor recognition of unorganized pictures even up to the fifth grade suggests a persistent difficulty in organizing unfamiliar material and relating it to accessible cognitive structures. This may require the use of more deliberate strategies, a task at which adults excel.

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